

Generation Interconnection System Impact Study Report

For Queue Project Q20-1566

James Black Tap 100 kV

February 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1566. The Interconnection Customer proposes a 210.6 MW battery energy storage facility interconnecting at the James Black Tap 100 kV substation in the Ohio area. The request was received on September 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 516 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$97,973,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
JAMES BLACK TP / SHERWOD FOREST 100 kV Circuit 1	SHERWOOD FOREST / JAMES BLACK TAP SUBSTATION 100 kV Circuit 2	152.7	162	57.2
SHERWOOD FOREST / JAMES BLACK TAP 100 kV #2	JAMES BLACK TAP / SHERWOOD FOREST 100 kV	148.5	167	57.2
SIBLEY NO TWO / GOLDTOWN 100 kV Line	SHERWOOD FOREST 230/1kV TX	146.9	112	21.2
MECHANICVILLE HYDROELECTRIC STA / SMITHERS 100 kV	SMITHERS / MECHNICVILLE HYDROELECTRIC STA 100 kV #2	135.8	187	10.6
MECHANICVILLE HYDROELECTRIC STA SUBSTATION / SMITHERS 100 kV	MECHANICVILLE HYDROELECTRIC STA / SMITHERS 100 kV #1	135.6	188	10.6
RIO BRAVO / SHERWOOD FOREST 100 kV line	FLATLICK 500/1 kV Transformer	126.9	163	21.5
PUERTA DE TIERRA / SPRINGFIELD - NICHOLS ST 100 kV	SHERWOOD FOREST 230 kV/1 kV Autotransformer	126.7	144	26.9
W LAFAYETTE SUBSTATION / MOSSYROCK 100 kV	SHERWOOD FOREST 230 kV/1 kV TX	124.1	147	7.2
UNKNOWN12262 / W LAFAYETTE 100 kV	SHERWOOD FOREST 230-1 kV Auto	120.4	145	7.2
UNKNOWN12262 / WEST LANCASTER 100 kV	SHERWOOD FOREST 230 kV/1 kV Transformer	118.5	144	7.2
MILL RUN / ELKVIEW COAL SUBSTATION 100 kV	SHERWOOD FOREST 230/1kV Transformer	118.1	167	26.9
SPRINGFIELD - NICHOLS ST / RIO BRAVO 100 kV line	SHERWOOD FOREST 230 kV/1 kV XFMR	118.0	143	26.9
SNIPES / JAMES BLACK TAP 100 kV Line 1	SNPES / JAMES BLACK TAP 100 kV Line 2	117.7	160	31.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
JAMES BLACK TAP / SNIPES 100 kV Cir. 2	JAMES BLACK TAP / SNIPES 100 kV Cir. 1	117.7	160	31.7
MILLRUN / PUERTA DE TIERRA 100 kV	SHERWOOD FOREST SUBSTATION 230/1kV Transformer	114.1	169	26.9
CATLETT / MECHANICVILLE HYDROELECTRIC STA 100 kV	MECHANICVILLE HYDROELECTRIC STA / BUTT VALLEY 100 kV	112.8	145	5.6
WEST LANCASTER / CHOUPIQUE 100 kV	SHERWOOD FOREST 230 kV/1 kV Autotransformer	102.2	133	7.2
BOYD / BUTT VLY 100 kV	SHERWOOD FOREST 230/1kV Autotransformer	101.7	113	26.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JAMES BLACK TP / SHERWOD FOREST 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$3,721,000	100% (\$3,721,000)
SHERWOOD FOREST SUBSTATION / JAMES BLACK TAP 100 kV	Rebuild line with higher-capacity conductor	\$5,403,000	100% (\$5,403,000)
GOLDTOWN / SIBLEY NO TWO 100 kV L	Rebuild line with higher-capacity conductor	\$4,490,000	100% (\$4,490,000)
MECHANICVILLE HYDROELECTRIC STATION / SMITHERS 100 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,809,000	100% (\$1,809,000)
MECHANICVILLE HYDROELECTRIC STA / SMITHERS 100 kV	Rebuild line with higher-capacity conductor	\$3,223,000	100% (\$3,223,000)
RIO BRAVO / SHERWOOD FOREST 100 kV line	Rebuild line with higher-capacity conductor	\$4,736,000	44% (\$2,091,000)
SPRINGFIELD - NICHOLS STREET / PUERTA DE TIERRA 100 kV Line	Rebuild line with higher-capacity conductor	\$7,839,000	100% (\$7,839,000)
MOSSYROCK / W LAFAYETTE 100 kV L	Rebuild line with higher-capacity conductor	\$9,884,000	100% (\$9,884,000)
W LAFAYETTE / UNKNOWN12262 100 kV Line	Rebuild line with higher-capacity conductor	\$4,807,000	100% (\$4,807,000)
UNKNOWN1222 / WEST LANCASTER 100 kV L	Rebuild line with higher-capacity conductor	\$5,806,000	100% (\$5,806,000)
MILL RUN / ELKVIEW COAL SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$3,550,000	100% (\$3,550,000)
SPRINGFIELD - NICHOLS ST / RIO BRAVO 100 kV line	Rebuild line with higher-capacity conductor	\$11,928,000	100% (\$11,928,000)
SNIPES SUBSTATION / JAMES BLACK TAP 100 kV	Rebuild line with higher-capacity conductor	\$4,492,000	100% (\$4,492,000)
SNIPES SUBSTATION / JAMES BLACK TAP 100 kV	Rebuild line with higher-capacity conductor	\$4,424,000	100% (\$4,424,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MILL RUN / PUERTA DE TIERRA 100 kV	Rebuild line with higher-capacity conductor	\$11,640,000	100% (\$11,640,000)
MECHANICVILLE HYDROELECTRIC STA / CATLETT 100 kV	Rebuild line with higher-capacity conductor	\$8,777,000	100% (\$8,777,000)
WEST LANCASTER / CHOUPIQUE 100 kV	Reconductor line	\$1,599,000	100% (\$1,599,000)
BUTT VALLEY / BOYD 100 kV	Reconductor line	\$2,490,000	100% (\$2,490,000)
	Total allocated to project		\$97,973,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.